

Colton Quirk

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EDUCATION

University of St Andrews

Master's of Science, Astrophysics

St Andrews, Scotland

Aug 2023 — May 2027

- Relevant Coursework:
- Second Semester Research: *Dark Matter in Early Type Galaxies*
 - Modeled the dark matter content of NGC ??? using ??? software packages
 - Developed a pipeline for the analysis of the dark matter content in a sample of galaxies
- Thesis: *Kinematic Signatures of Spiral Shocks*
 - Developed a Julia package, `SPHAnalysis`, to work with Smoothed Package Hydrodynamic (SPH) galaxy simulation data
 - Analyzed data from an SPH galaxy simulation to identify signatures of shocking regions within spiral arms of galaxies

University of Arizona

Bachelor's of Science, Physics and Astronomy, Minor: Computer Science

Tucson, AZ

Aug 2023 — May 2027

- Cumulative GPA: 3.98/4.0 | Dean's List with Distinction, National Merit Scholarship
- Relevant Coursework:
- Honors Thesis: *Scientific Visualization of Photon Ring Formation and Imaging*
 - Developed a custom rendering engine for Blender to view models around a Schwarzschild Black Hole
 - Created a movie demonstrating the process of “imaging” a black hole based on the BHEX mission

WORK EXPERIENCE

National Radio Astronomy Observatory (NRAO) REU

May 2024 — Present

National Radio Astronomy Observatory (NRAO)

Socorro, NM

- *Probing Radio Environments of Superclusters in the Very Large Array Sky Survey (VLASS)*
- Convinced high-performance computers to work overtime without unions, reducing simulation time by 50%
- Wowed a room full of nerds with pretty pictures of invisible things and imaginary findings

Arizona Space Grant Consortium Internship

Dec 2023 — Mar 2024

University of Arizona, Arizona Space Grant Consortium

Tucson, AZ

- *Revealing the Far-Ultraviolet Background Radiation through Archival Astronomical Data*
- Developed HIPAA-compliant digital signatures, because doctors' handwriting wasn't illegible enough already
- Turned spaghetti code into a gourmet dish, making other interns drool with envy

Telescope Operator

Jun 2020 — May 2023

The University of Arizona, Steward Observatory

Tucson, AZ

- Scaled user base from 10 to 2000+, accidentally becoming a small wealthy nation in the process
- Crafted Bash scripts so clever they occasionally made other engineers weep with joy
- Automated support responses, reducing human interaction to a level that would make introverts proud
- Built a documentation site that actually got read, breaking the ancient RTFM curse

PROJECTS

EXTRACURRICULAR ACTIVITIES

SKILLS

- **Programming Languages:** Julia, Python, HTML/CSS, Java, Javascript, Fortran
- **Technologies:** Git, UNIX, LaTeX, Quarto, Typst, CASA, Blender
- **Languages:** English, Spanish